MIL-C-48023 (PA)
6 August 1973
SUPERSEDING
(See 6.9)

MILITARY SPECIFICATION

CARTRIDGE, 107MM, HE: M453

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Cartridge designated as Cartridge, 107MM, HE: M453.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for

PURCHASE DESCRIPTIONS

ARMY

PA-PD-2566 - Cartridge, Ignition, M2A2, Parts for Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Armament Command

F-9211934	- Cartridge, 107MM, HE, M453
C-8864766	- Box, Packing, Ammunition for 107MM, Mortar Ammunition in Fiber Container, M507
F-8864767	- Container, Ammunition, Fiber, M507, for 107MM Mortar Ammunition
C-8841144	- Torque Fixture
D-8841397	- Pneumatic Vise

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

. Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 18-67 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge.-The cartridge and components shall comply with all requirements specified on Drawing (Dwg.) F-9211934, all associated drawings, and with all requirements specified in applicable specificatio

3.3 Air pressure test of cartridge.-The loaded cartridge shall withstand an air pressure of five (5) to five and one quarter (5.25) pounds per square inch gage (psig) for ten (10) seconds, minimum(min) without a drop in pressure when tested as specified in 4.4.4.

3.4 Ballistic functioning.

3.4.1 Cartridges.-The cartridge shall function satisfactorily (See 6.5.1) when tested as specified in 4.4.5 (Not applicable to safety phase).

3.4.2 Grenade.-The grenades shall function satisfactorily (see 6.5.2) and shall not detonate in the air stream upon ejection from the projectile when tested as specified in 4.4.5. (Not applicable to safety phase).

3.4.3 Safety.-There shall be no premature burst (see 6.5.5), or evidence thereof, in any of the test firings. In addition, there shall be no premature burst or separation of the metal parts in the gun bore or in flight when the cartridge is fired using a propelling charge adjusted to give 112 percent of the average pressure produced by the propellant loaded in the particular ammunition lot. Testing shall be as specified in 4.4.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

3.4.4 Range.-The range of any one cartridge shall be not less than ninety (90) percent of the average range of all cartridges fired in any one phase, using, approximately, the same elevation of the weapon and the same fuze setting time. Testing shall be as specified in 4.4.5. (Not applicable to safety phase). (see 6.8).

3.5 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.6)

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.
f. Number of units of product inspected.
g. Results obtained, by code number, for all inspections performed.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1.) The first article sample shall consist of the following:

a. Seventy-five (75) groove pins.
b. Cartridges (as specified in 4.2.2.2)

All parts and completed assemblies submitted shall have been produced by the Contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used drawing specification or source of supply as to significantly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, the following tests shall be performed:

4.2.2.1 Hardness of groove pin (See Dwg. B-9211940) Code No. 01001.-The seventy-five (75) groove pins shall be subjected to this test as specified in 4.4.2.

4.2.2.2 Ballistic functioning (See 3.4).-These tests shall be conducted on the sample quantities specified herein beginning with the first lot produced and continuing until three (3) consecutive lots have passed the acceptance criteria specified. Thereafter, sampling plans for these tests shall be in accordance with 4.3.3.6. (Regular production). The Government inspector shall select fifteen (15) cartridges from each lot. The contractor shall forward these cartridges to a Government-owned proving ground designated by the Contracting Officer for ballistic functioning tests as specified in 4.4.5.

4.2.2.2.1 Rejection.-The cartridge lot shall be rejected if any one of the following occurs:

a. Two (2) or more cartridges fail to function (including fuze malfunctions). (See 6.5.1) Code No. 02001.
b. The number of grenades failing to function equals or exceeds sixteen (16) (not applicable to safety phase and excluding) in-housing bursts and air bursts) (See NOTE below and 6.5.2) Code No. 02002.
c. The number of grenade in-housing burst equals or exceeds thirty-two (32) (not applicable to safety phase) (See 6.5.4) Code No. 02003.
d. Grenades fired for range fail to comply with the applicable range requirement. Code No. 02004.
e. Any premature burst occurs. (See 6.5.5) Code No. 02005.
f. Any separation of metal parts occurs prior to initiation by time fuze. Code No. 02006.
g. The number of grenade air bursts equals or exceeds nine (9) (See 6.5.3) Code No. 02007.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

NOTE: If one (1) cartridge functions below 1,000 feet, an additional cartridge will not be required, provided the total score of all defects does not exceed the prescribed acceptance criteria. If the grenade acceptance criteria are exceeded, the results of that cartridge shall not be considered for grenade functioning and an additional cartridge shall be selected for firing in the applicable phase for grenade functioning acceptance from those cartridges designated for the safety phase firing.

4.2.2.2.2 Retest.-If, for any reason, the proof director considers that the test conditions have detrimentally affected the test results, he may declare the results invalid and perform a new test on additional sample cartridges as required.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any part or assembly, in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Change to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Cartridge metal parts assemblies of one lot interfix number from one manufacturer.
b. Loaded fuzes of one lot interfix number from one manufacturer.
c. Grenade housing assemblies of one lot interfix number from one manufacturer.
d. Loaded grenades of one lot interfix number from one manufacturer. See Not)
e. Propelling charge assemblies of one lot interfix number from one manufacturer.
f. Loaded ignition cartridges from not more than two lots. (See 4.3.3.5)
g. Obturating mechanism parts and assemblies of one lot interfix number from one manufacturer.
h. Charge propellant of one lot from one manufacturer.

NOTE: Only those grenade lots which have successfully completed the functioning test indicated in the specification covering the loading of the grenade shall be assembled into the body assembly.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plan for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Pin, twist (See Dwg. B-9211942).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Diameter, minimum (min)	Gage	03001
102.	Length, min	Gage	03002
Minor			
201.	Protective coating damaged (base metal exposed)	Visual	03003
202.	Evidence of poor workmanship (See 3.5)	Visual	03004

4.3.2.2 Charge assembly, prior to assembly of cover (See Dwg. C-9211943).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Weight of charge	Balance	04001
102.	Assembly damaged (puncture, cut or dent)	Visual	04002
Minor			
201.	Evidence of poor workmanship (See 3.5)	Visual	04003

4.3.2.3 Grenade stackup (upon insertion of each grenade layer into body (See Dwg. F-9211934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Safety wire not removed from each grenade	Visual	05001
102.	Grenade missing or improperly positioned	Visual	05002
103.	Grenade vane damaged	Visual	05003
104.	Liner missing	Visual	05004
105.	O-ring missing or improperly seated on forward plate	Visual	05005
106.	Red dot not visible on housing of each grenade	Visual	05006

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Minor			
201.	Evidence of poor workmanship (See 3.5)	Visual	05007

4.3.2.4 Base plug, prior to assembly to cartridge (See Dwg. F-9211934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	O-ring missing	Visual	06001
102.	Threads damaged	Visual	06002
Minor			
201.	Evidence of poor workmanship (See 3.5)	Visual	06003
202.	Grease missing from o-ring	Visual	06004

4.3.2.5 Cartridge, prior to assembly of groove pins and twist pins (See Dwg. F-9211934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Diameter of groove pin holes	Gage	07001
102.	Diameter of twist pin holes	Gage	07002
103.	Depth of twist pin holes	Gage	07003
104.	Location of twist pin holes	Gage	07004
105.	Groove and twist pin holes overlap or interface (holes insufficiently separated)	Visual	07005
Minor			
201.	Evidence of poor workmanship (See 3.5)	Visual	07006

4.3.2.6 Fuze assembly (with charge assembly and O-ring) (See Dwg. F-9211943).-

Categories	Defects	Method of Inspection	Code No.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Critical: None defined

Major			
101.	Length to fuze thread shoulder, max	Gage	08001
102.	Charge assembly cap damaged (punctured, cracked or dented)	Visual	08002
103.	O-ring missing from fuze shoulder	Visual	08003
104.	Cap stake missing (4) or incorrect	Visual	08004
Minor			
201.	Total length, max	Gage	08005
202.	Evidence of poor workmanship (See 3.5)	Visual	08006
203.	Grease missing from O-ring of fuze assembly	Visual	08007
204.	Fuze thread damaged	Visual	08008

4.3.2.7 Cartridge, prior to assembly of fuze and obturating mechanism components (See Dwg. F-9211934).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Groove pin missing	Visual	09001
2.	Twist pin missing	Visual	09002
Major			
101.	Gap between base plug and body max	Gage	09003
102.	Groove pin above flush	Visual/Manual	09004
103.	Twist pin above flush	Visual/Manual	09005
104.	Forward plate missing or damaged	Visual	09006
Minor			
201.	Evidence of poor workmanship in expelling charge assembly cavity (See 3.5)	Visual	09007
202.	Marred surface not recoated	Visual	09008

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.3.2.8 Cartridge, less propellant holders and increments (See Dwg. F-9211934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Rotating disc fails to withstand disassembly torque	Test	10001
102.	Cartridge container fails to withstand disassembly torque	Test	10002
103.	Pressure plate nut fails to withstand disassembly torque	Test	10003
104.	Striker nut assembly fails to withstand diassembly torque	Test	10004
105.	Fuze fails to withstand disassembly torque	C8841144 D8841397	10005
106.	Pressure plate of obturating mechanism reversed (beveled edge not in rotating disc)	Visual	10006
107.	Pin at end of obturating mechanism missing or improperly positioned	Visual	10007
108.	Fuze staked improperly (2 places)	Visual	10008
109.	Any obturating mechanism part improperly assembled or damaged	Visual	10009
110.	Ignition cartridge missing (when viewed through holes in obturating mechanism)	Visual	10010
111.	Pressure plate nut stake (2) missing or incorrect	Visual	10011
Minor			
201.	Total length, max	Gage	10012
202.	Protective coating with bare spot	Visual	10013
203.	Marking misleading or unidentifiable including incorrect color code	Visual	10014
204.	Evidence of poor workmanship (See 3.5)	Visual	10015

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

205.	Fuze assembly damaged (dented or deformed)	Visual	10016
206.	Fuze not set on safe position	Visual	10017

4.3.2.9 Cartridge (See Dwg. F-9211934).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Outside diameter	Gage	11001
102.	Any propellant increment bundle missing, insecure or damaged (cut or torn)	Visual/Manual	11002
103.	Any propellant holder missing or incorrectly positioned	Visual	11003
Minor			
201.	Evidence of poor workmanship (See 3.5)	Visual	11004

4.3.2.10 Unsealed fiber container (See Dwg. F-8864767).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Support ring missing	Visual	12001
102.	Support tube missing	Visual	12002
Minor			
201.	Cartridge not removable by hand from container	Manual	12003
202.	Filler missing	Visual	12004
203.	Cushion missing	Visual	12005

4.3.2.11 Sealed fiber container (See Dwg. F-8864767).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Major			
101.	Metal end loose or distorted	Visual/Manual	13001
102.	Container cut or damaged through all impregnated layers	Visual	13002
103.	Tape incomplete or badly wrinkled	Visual	13003
Minor			
201.	Contents loose	Manual	13004
202.	Gap between cover and body of container exceeds 1/8 inch	Visual	13005
203.	Marking misleading or unidentifiable including incorrect color code	Visual	13006

4.3.2.12 Sealed wooden packing box (See Dwg. C-8864766).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Box damaged	Visual	14001
102.	Strapping loose or broken	Visual	14002
103.	FSN designation incorrect	Visual	14003
Minor			
201.	Strapping improperly assembled	Visual	14004
202.	Car seal missing, unsealed, or improperly positioned	Visual	14005
203.	Marking misleading or unidentifiable, including incorrect color code	Visual	14006

4.3.3 Testing.-

4.3.3.1 Weight of cartridge (information use only) (See Dwg. F-9211934) Code No. 15001.-This determination shall be performed 100 percent using the procedure specified in 4.4.1.

4.3.3.2 Hardness of groove pin (See Dwg. B-9211940) Major Defect, Code No. 16001.-Twenty (20) groove pins shall be selected from each heat treatment batch (See Note 1) after completion of the heat treatment process and subjected to this test. If one or more groove pins fail to comply with the specified hardness

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

requirement, the quantity represented by the failing groove pin shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

NOTE 1 - A heat treatment batch shall be defined as parts that have been heat treated at the same time in the same furnace and quench bath for each of all phases of the entire heat treatment process.

4.3.3.3 Moisture content of charge propellant, (See Dwg. C-9211943) Major Defect, Code No. 17001.-The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture content of each lot of propellant shall be determined at the loading station at the time of loading the charge cups. This determination shall be made at least once each eight (8) hours of production. Sampling of the propellant shall be in accordance with MIL-STD-286, Method 101.1.2. In addition, if the moisture content fails to comply with the requirement specified on the applicable drawings and loading has not begun, the lot shall be rejected until corrective action has been taken as verified by repeating the moisture content determination. If loading has begun, charge assemblies loaded with the non-conforming propellant shall be rejected. This determination shall be performed as specified in 4.4.3.

4.3.3.4 Air pressure test of cartridge, (See 3.3) Major Defect, Code No. 18001.-Eighty (80) loaded cartridges, with fuze, shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4. If two (2) or more cartridges fail to comply with the applicable requirement, the lot shall be rejected.

4.3.3.5 Check test for deterioration of ignition cartridge (See PA-PD-2566) Major Defect, Code No. 19001.-If the total time between the original acceptance of the ignition cartridge lot and the assembly of that lot into complete round cartridges exceeds two (2) years, or if the ignition cartridges have been subjected to adverse conditions, however brief, at any time since previous tests, the ignition cartridge lot shall be subjected to, and must satisfactorily pass the check test for deterioration specified in PA-PD-2566 immediately before the ignition cartridge lot is assembled into the complete round cartridge. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges. (See 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.3.3.6 Ballistic functioning (See 3.4 and Dwg. F-9211934) (Regular Production).-

4.3.3.6.1 Sampling.-Following successful completion of first article testing, the Government inspector shall select five (5) cartridges from each lot for proving ground acceptance testing. When fifteen (15) cartridges have been accumulated, they shall be forwarded to the proving ground for test as specified herein. The tests shall be performed as specified in 4.4.5.

4.3.3.6.2 Rejection.-The three (3) lots shall be rejected if any one of the following occur:

a. Two (2) or more cartridges fail to function (including fuze malfunctions). (See 6.5.1) Code No. 20001.
b. The number of grenades failing to function equals or exceeds sixteen (16) (not applicable to safety phase and excluding in-housing bursts and air bursts) (See NOTE below and 6.5.2) Code No. 20002.
c. The number of grenade in-housing bursts equals or exceeds thirty-two (32) (notapplicable to safety phase) (See 6.5.4) Code No. 20003.
d. Grenades fired for range fail to comply with the applicable range requirements. Code No. 20004.
e. Any premature burst occurs (See 6.5.5) Code No. 20005.
f. Any separation of metal parts occurs prior to initiation by time fuze Code No. 20006.
g. The number of grenade air bursts equals or exceeds nine (9) (See 6.5.3) Code No. 20007.

NOTE: If one (1) cartridge functions below 1,000 feet, an additional cartridge will not be required, provided the total score of all defects does not exceed the prescribed acceptance criteria. If the grenade acceptance criteria are exceeded, the results of that cartridge shall not be considered for grenade functiong and an additional cartridge shall be selected for firing in the applicable phase for grenade functioning acceptance from those cartridges designated for the safety phase firing.

4.3.3.6.3 Retest.-If, for any reason, the proof director considers that the test conditions have detrimentally affected the test results, he may declare the results invalid and perform a new test on additional sample cartridges as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in 4.3.3 (See 6.3)

4.4 Test methods and procedures.

4.4.1 Weight of cartridge.-The loaded projectile, less the obturating mechanism components and fuze and expelling charge assembly, shall be weighed, using methods and equipment as specified. All weights shall be recorded for informational use.

4.4.2 Hardness of groove pin.-The groove pins shall be subjected to the hardness test in accordance with ASTM E 18-67. (Part 31) This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.3 Moisture content of charge propellant.-The moisture content of the charge propellant shall be determined at the time and place of loading and in accordance with MIL-STD-286, Method 101.1.2.

4.4.4 Air pressure test of cartridge.-The loaded cartridge, with fuze, shall be assembled to a fixture as shown on the equipment list and a measured quantity of air applied to produce the air pressure as specified in the applicable requirement. Observation shall be made for evidence of leakage.

4.4.5 Ballistic functioning.-These tests shall be performed at a Government-owned proving ground in accordance with the applicable Ordnance Proof Manual and Acceptance Test Procedure. Cartridges shall not be fired for acceptance testing when the wind velocity, at an altitude of 1000 to 1400 feet, exceeds twenty-five (25) miles (21.7 knots) per hour. The cartridge shall be fired in a weapon for which it is designated at a quadrant elevation and with a fuze setting that will cause ejection of the cargo at a minimum height of one thousand feet above a prepared impact field. Determination shall be made for conformance to applicable requirements. In addition, to other information recorded, a record shall be made of the following: Fuze duds, grenade duds, and off-time fuze functioning. Ballistic data from all firings shall be recorded and forwarded as specified in 6.7. Each cartridge lot shall be represented by at least one cartridge in each of the following tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.4.5.1 Seven-increment charge.-Five (5) of the sample cartridges shall be subjected to this test with fuze setting as specified in 4.4.5, and the appropriate seven-increment charge.

The center of impact of each group of grenades shall be determined and recorded for each round fired at this charge. When requested, equipment shall be provided to measure velocities of projectiles fired in this test with velocity results recorded.

4.4.5.2 Maximum-increment charge (as specified on applicable drawing).-Five (5) of the sample cartridges shall be subjected to this test with fuze setting as specified in 4.4.5 and the maximum increment charge. The center of impact of each group of grenades shall be determined and recorded for each round fired at this charge.

4.4.5.3 Safety.-Five (5) of the sample cartridges shall be subjected to this firing, using a propelling charge adjusted to give 112 percent of the average pressure produced by the propellant used in the particular ammunition lot. (This average value of pressure appears on the data card for each propelling charge.) Observation shall be made for premature bursts and separation of metal parts in the gun bore or while in flight. (See 6.7)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with Dwg. F-8864767.

5.2 Packing.

5.2.1 Level A.-The cartridges, preserved and packaged as specified in 5.1.1, shall be packed in accordance with Dwg. C-8864766.

5.3 Marking.-Marking shall be in accordance with Dwgs. C-8864766 and F-8864767 and Code of Federal Regulations, Title 49 Parts 171-179 and Army Regulation AR 380-55.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423)

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricatio or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of check test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the ignition cartridges. The tests shall be conducted at Government expense without cost to the contractor assembling the ignition cartridges onto the complete round, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the ignition cartridge lot was under the jurisdiction of either contractor.

6.5 Cartridge functioning and grenade functioning.-

6.5.1 Cartridge.-Satisfactory functioning of the cartridge is defined as ejection of the grenades fromthe cartridge at a pre-determined time after firing.

6.5.2 Grenades.-Satisfactory functioning of the grenade is defined as detonation of the ball assembly in the air after ground impact and ejection from its housing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

6.5.3 Airburst.-A grenade air burst is defined as functioning of the grenade in the air soon after ejection from the body assembly and prior to ground impact.

6.5.4 Burst In-housing.-A grenade burst-in-housing (BIH) is defined as high order functioning of the ball assembly in its housing upon ground impact.

6.5.5 Premature burst.-A premature burst is considered to have occurred if the cartridge functions in the gun bore, or if the time of flight is less than 1.5 seconds.

6.6 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army Munitions Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, Attn: SARA-QA-A-S.

6.7 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.8 Range .

6.8.1 Average range.-The average range of all cartridges fired in any one phase is computed by averaging the horizontal distances from the weapon from which the cartridges were fired to the determined center of impact of grenades for each of the cartridges.

6.8.2 Center of impact.-The center of impact of grenades, as used in this specification, is defined as the point around which grenades, ejected from any one cartridge, impacted the ground.

6.9 Supersession data.-This specification includes the requirements of, and supersedes; Purchase Specification AEI-392, dated 20 February 1968.

6.10 Distribution of ammunition data cards.-Distribution of data cards shall include the following: Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S, Dover, New Jersey 07801.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1315-A005

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.